

Creating an app

In this article, we are going to create a torch app for an Android mobile using Apache Cordova. This is an app that will switch the camera flash of the mobile, on or off, on the press of a button.

Let's start by going to the directory in which you want to create your Cordova project (in my case, it's C:\project) via the command prompt (if you are using Windows OS) or the terminal (if you are using Linux OS), and run the following command:

```
C:\project>cordova create torchapp com.app.torch Torch
```

This command statement creates the required directory structure for your Cordova app.

Here is a brief description.

- **cordova create:** This script generates a default Web-based application whose home page is the *index.html* file in */www* directory. This page loads up when you run your app. In short, this is the entry point of the app.
- **torchapp:** This is the name of the directory in which the app will be created.
- **com.app.torch:** This is the default reverse domain value, in which the classes will be stored. You can use your own domain value.
- **Torch:** This is the name (title) of the app.

After running the above command, a new Cordova project will be created in the *torchapp* directory, which will have the following structure:

```
.hooks\
. platform\
. plugins\
. www\
o css\
o img\
o js\
o index.html
. config.xml
```

- **hooks:** This directory is used to store scripts for customising *cordova-cli* commands.
- **platform:** This directory contains all the scripts and source code for the platform that you add to your Cordova project.
- **plugins:** This directory is used to store plugins, which will be used in the app.
- **www:** This directory stores all the Web artefacts of the project, such as CSS, HTML and JavaScript files. Most of the code will be stored here. In short, this is the brain of the Cordova app.
- **config.xml:** This file contains all the important information about the Cordova app, such as its name, description, content-src, etc. It allows you to customise the behaviour of your project.

Adding the platform

Since we are creating an app for the Android platform, we need to add the platform to the Cordova project. Go to the Cordova project's directory, *torchapp*, and run the following command:

```
C:\project\torchapp>cordova platform add android -save
```

The above command statement will add the Android platform to the Cordova project. If we want to add any other platform, say iOS, we just need to replace the keyword 'android' with 'iOS' and run the command. The above command statement will also create the important files required for the Android platform in the app's *platform* directory.

Prerequisites for building an app

To build and run apps on the computer, you need to install SDKs for each targeted platform; or if you are using a browser for the development, you can use a browser platform that does not require any platform SDKs.

You can check if your system meets the requirements for building the platform by running the following command:

```
C:\project\torchapp>cordova requirements
```

Since we are targeting the Android platform for the app, it will give the following as the output:

```
Requirements check results for android:
Java JDK: installed
Android SDK: installed
Android target: installed android-21,android-22,android-23,Google Inc.:Google APIs:22
Gradle: installed
```

This check is necessary to make sure we have installed all dependencies.

Creating the app

We are now going to create the app for an Android mobile phone; we have already created directories and other important files for this app.

Technologies we are going to use are:

- HTML – for the user interface design
- CSS – for styling the HTML page
- JavaScript – for the logic of the app
- Flashlight-PhoneGap-Plugin – to access the mobile camera's flash

Creating the app interface

First, we need some icons/images for the application. Visit <http://www.flaticon.com> and download the icon of the power button, preferably of size 64x64 pixels in .png format, rename